ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT CODE: 0612 451 07A
CAT 3
Total marks: 40 Time: 1 hour 30 minutes

Name: _____ Reg. No: _____ Date: _____

Instructions to Candidates

- ✓ Answer ALL questions.
- ✓ Marks are shown in brackets at the end of each question or part.
- ✓ Write clearly and show your working where a calculation is required.
- ✓ Mobile phones and unauthorised materials are not allowed in the examination room.

QUESTIONS

1. Define network testing. **(2 marks)**

2. State two reasons why network testing is important. **(2 marks)**

3. Name the instrument used to measure current without cutting or disconnecting the wire, and state how a voltmeter is connected to the component being tested. **(2 marks)**

4. State the role of Npcap when using Wireshark on a Windows computer. **(2 marks)**

5. Define a VLAN. **(2 marks)**

Question 6 (6 marks)

(a) Define TTL and state what a router does when the TTL of a packet reaches 0. **(2 marks)**

(b) A Windows computer sends a packet with a starting TTL of 128. The packet passes through 9 routers before reaching its destination. Calculate the TTL on arrival and state whether the packet is dropped. **(2 marks)**

(c) Explain how the tracert command uses TTL to discover each router on the path to a destination. **(2 marks)**

Question 7 (6 marks)

A technician runs tracert to a web server and obtains the output below.

```
1    2 ms    1 ms    2 ms   192.168.1.1
2    9 ms    8 ms    9 ms   10.20.0.1
3   26 ms   25 ms   27 ms   41.89.10.5
4  180 ms  178 ms  182 ms   196.201.4.9
5    *       *       *      Request timed out.
6  190 ms  188 ms  191 ms   203.0.113.20
```

(a) Identify the hop where the delay increases sharply and state what this suggests about the network. **(2 marks)**

(b) Give two possible reasons for the asterisks (* * *) shown at hop 5, even though hop 6 replied. **(2 marks)**

(c) Explain why running ping alone would not have shown where the delay occurs. **(2 marks)**

Question 8 (6 marks)

(a) Describe the TCP three-way handshake as seen in Wireshark, stating who sends each packet and the SYN and ACK flag values in each step. **(4 marks)**

(b) Write the Wireshark display filter that would show only:

(i) the first packet (SYN) of each TCP connection request; **(1 mark)**

(ii) ICMP Echo Request packets. **(1 mark)**

Question 9 (6 marks)

A technician is commissioning a new office network and records the three findings below.

Finding	Tool used	Observation
(i)	Voltmeter	A wall socket reads 180 V AC. The expected supply is about 240 V AC.
(ii)	Cable tester	A newly crimped T568B patch cord shows LEDs 1 to 4 and 6 to 8 lighting in order, but LED 5 does not light.
(iii)	Wi-Fi analyser	The signal in the meeting room is measured at -82 dBm.

For each finding, interpret the result and recommend one suitable action. **(2 marks each)**

Question 10 (6 marks)

A school has a Finance department (PC1 and PC2 on ports Fa0/1 and Fa0/2) and a Human Resource department (PC3 and PC4 on ports Fa0/3 and Fa0/4) connected to one Cisco switch. The Finance network is 192.168.10.0/24 and the Human Resource network is 192.168.20.0/24. The school wants the two departments separated.

(a) Write the switch commands to create VLAN 10 named FINANCE and to assign port Fa0/1 to it as an access port. **(3 marks)**

(b) After the VLANs are configured, PC1 (192.168.10.10) cannot ping PC3 (192.168.20.10). Explain why, and describe how communication between the two VLANs can be enabled. **(3 marks)**

END OF QUESTION PAPER